Palak Deb Patra

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EXPERIENCE

NatWest Group · Senior Analyst / Software Engineer July 2025 – Present

- Build backend systems, APIs, and data workflows for enterprise applications.

NatWest Group · Software Engineering Intern May – July 2024 · Gurgaon, India

- Collected organization-wide DevOps metrics using GitLab GraphQL and REST APIs.
- Automated paginated data retrieval and CSV exports in Python, reducing processing time by 28%.
- Unified 16 CSV datasets in Snowflake and connected the data to ThoughtSpot dashboards.

SELECTED PROJECTS

The Engineering Atlas HTML · CSS · JavaScript

Interactive engineering courses with simulations, quizzes, and interview practice.

- Nine engineering subjects, with learning paths, progress tracking, and downloadable offline courses.

CodeShift Python · LangGraph · TypeScript · Ollama

Translate Python modules to TypeScript and verify their behavior.

- One recorded run verified 15 of 31 modules as equivalent and exercised 197 functions and methods.

Memex Python · LangGraph · ChromaDB · SQLite · Ollama · Streamlit

Search your documents and get answers with source citations.

- Combines semantic and keyword search with local answers, search filters, and retrieval analytics.

FrameSmith Python · LangGraph · scikit-optimize · PresentMon · NVML · SQLite

A prototype for exploring graphics settings with GPU telemetry and optimization.

- Records settings, benchmark trials, and scores for inspection; graphics improvements remain unvalidated.

PUBLICATION

Bidirectional Convolutional-GRU Integrated GAN for Unpaired Low-Light Image Enhancement

First author · IEEE SPCOM 2026 · July 2026 · Pages 1–5

A GAN integrates bidirectional convolutional GRU modules to capture local textures and longer-range dependencies, learning from unpaired images to enhance low-light detail and structure.

Palak Deb Patra, Santosh Kumar Panda, Manoj Kumar Bishwal, and Pankaj Kumar Sa · DOI

EDUCATION

NIT Rourkela · Computer Science & Engineering December 2021 – June 2025

CGPA: 8.52 / 10

SKILLS

Programming Languages	Python · Java · C++ · JavaScript · TypeScript · SQL
Databases	Snowflake · Oracle · MySQL · ChromaDB
AI/ML	PyTorch · TensorFlow · scikit-learn · OpenCV · spaCy · NLTK · NVIDIA CUDA
Agent Systems	LangGraph · LangChain · Ollama
Web Interfaces	React · Angular · Streamlit
APIs	FastAPI · REST APIs · GraphQL
Cloud	AWS (S3, EC2, RDS)
Development	Git · GitHub · Jupyter

ACTIVITIES & ACHIEVEMENTS

- **Competitive programming** — Solved 800+ LeetCode problems; peak rating 1795, top 8% globally. Global rank 277 in Weekly Contest 438; rank 1839 in Weekly Contest 437.
- **Amazon ML Summer School** (2023) — Selected for training in advanced machine learning topics.
- **NITRUTSAV · Event Management Coordinator** — Led seven volunteers for NIT Rourkela's cultural festival. Helped increase sponsorship by 20% and participant engagement by 11.26%.
- **AASRA · Senior Staff Member** — Coordinated outreach and fundraising for AASRA's DIYA Project. Contributed to team fundraising that raised ₹2,11,000 and generated ₹1,32,000 in profit.
- **LLMs & Prompt Patterns** (Vanderbilt University) — Completed the program with a perfect score of 100%.